About three weeks before the competition, the robots came up with this walk that was suddenly able to beat anything the other teams had.

The walk gave the UT Austin Villa players such an enormous advantage, says Stone, that he and his co-programmers were able to devote a considerable amount of each game to testing a pass-based strategy they knew would be less effective than the dribble-based strategy with which their robots scored most of their goals.

“We were thinking about next year,” says Stone. “If other teams catch up to us on the walk, which I think they will, then it’ll be much more about passing and positioning.”

That kind of rapid advancement, says Stone, is the point of the “Simulation League,” which is the only one of the five leagues in the tournament that doesn’t involve real nut-and-bolt robots. Instead, the games are played by two teams of nine autonomous artificial intelligence (AI) programs. The programs battle it out in a videogame-like environment.

Each “player” has to react, in real time, to the data being sent to it by its teammates and by the simulator, which models the physics of the real world and is constantly recalculating its datastream based on where the 18 different players are going, when they’re bumping into each other, when they’re falling down and where they kick the ball.

“It’s one level abstracted away from the real world,” says Stone. “The simulated robots don’t have a vision sensor. Instead they’re told their distance and angle from the ball and the goal, which is something you get from real robots only after some processing. They still have to figure out where they are on the field, however, and where their teammates are by combining data from each other. You still see them falling over when they get knocked into each other. They still have to learn how to walk, which is the biggest technical challenge in the real and simulation leagues.”

Because it’s not nearly as expensive to test simulated robots, and because much of the experimentation can be done in the wee hours, without humans around to pick up the robots that fall down, there’s more time and less cost to trying out new strategies. And because RoboCupSoccer is fundamentally an academic enterprise, dedicated to advancing science, the novel strategies that succeed are published in academic journals, inspiring other teams to incorporate the advances.

RoboCup has already changed the world, says Stone. Robots first developed for the competition were recently deployed in Fukushima, Japan, after the nuclear reactor disaster. One of the early winners of